CURRICULUM VITAE

# Muhammad Abu Bakar

Geospatial Data Specialist

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## Professional experience

11/2025 – present

**Geospatial Data Specialist** · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Open-source drone LiDAR workflow for power lines: georeferencing and classification of the raw point clouds, automated vegetation-to-conductor clearance analysis with risk categories; tree-fall risk analysis in development.
- Right-of-way and construction-site monitoring along pipelines: open-source georeferencing of RGB images via learned feature matching, object detection with a fine-tuned foundation model and an open-source VLM, the VLM also acting as gatekeeper for every detection.
- Live analysis: heavy-machinery detection on an NVIDIA Jetson, with detections sent directly to the customer.
- Backend with PostGIS and TiTiler for serving large GeoTIFFs.
- Drone flight planning: primary and duplicate checks.

03/2024 – 10/2025

**Geospatial (EO) Data Scientist, part-time** · Karuna Technology UG, Germany

- Differential InSAR pipeline with ISCE: automated acquisition, preprocessing and interferogram generation.
- Google Earth Engine workflows for Sentinel-1 time series (Refined Lee filter) feeding a soil-moisture ML model.
- Urban tree-species classification from fused multispectral and SAR data; training data for tree-crown segmentation.
- Vector data pre- and post-processing in QGIS.

04/2021 – 02/2024

**GIS & RS Specialist** · MapTransfer, Germany

- Spatial data workflows for a construction company: georeferencing, orthophoto mosaics, cloud-optimized GeoTIFFs on AWS, PostGIS–QGIS integration.
- In-house courses on LiDAR in QGIS and spatial data analysis; course development for Spatial SQL, ALKIS and GeoServer.
- Geodata management and visualization for a telecom project based on ALKIS.
- Web GIS: PostGIS vector layers in OpenLayers.

## Education

2017 – 2020

**MSc Geology** · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020

**MSc Remote Sensing & Geographic Information System** · National University of Sciences and Technology, Pakistan

- Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.

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| 2012 – 2015 | **BSc Geology** · Bahria University, Pakistan<br>○ Thesis: Petrophysical analysis of Joya Mair – Well 4. |

## Academic projects

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| Landsat · Sentinel-2 | Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000. |
| MODIS · TRMM | Snow cover and precipitation variability, 2000–2015. |
| Landsat 7 | Snow-cover change in the Kabul Basin and its impact on water resources. |
| Landsat 8 · ASTER | Land cover, erosion and morphometry of Mauna Kea. |

## Further training

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| 2022 | NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring.<br>UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy). |

## Skills

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| GIS | ArcGIS Pro, QGIS, ArcMap |
| Remote sensing | Multispectral, hyperspectral and SAR data; ENVI, ERDAS Imagine, eCognition, ESA SNAP |
| LiDAR | CloudCompare, WhiteboxTools; DEM and DSM generation |
| Cloud | Google Earth Engine, AWS (EC2, S3) |
| Databases | PostgreSQL/PostGIS, SpatiaLite, DuckDB, SQL |
| Programming | Python (GDAL, Fiona, Rasterio, scikit-learn, TensorFlow, Cartopy), R, Spatial SQL, Bash |
| Machine learning | CatBoost, CNNs, fine-tuned foundation models, VLMs |
| Web mapping | Leaflet, OpenLayers, GeoServer, TiTiler |
| Other | Git, LaTeX, Microsoft Office |

## Languages

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| English | C2 |
| German | B2 |

Freiburg im Breisgau, 14 October 2026

Signature